

Applied Data Analysis (CS401)



Lecture 14 ADA in action 17 Dec 2025

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EPFL

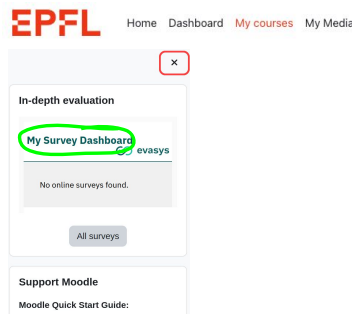
1

Announcements

- Today: last lecture [click [here](#) for a famous last lecture]
- No lab session this Friday
- Final project milestone P3 due on **Sun 21 Dec 2025**
- Final exam: **Tue 13 Jan 2026**, 15:15–18:15
 - Dry run, QA example released this week
 - Announcement re: exam protocol, room assignment, etc., to be made in early January

Course evaluation

Course evaluation
available on Moodle!



Today we will

show how everything you've
learned in lectures 1–13 comes
together in one project

Paper available at <https://doi.org/10.1073/pnas.2106152118>

3



Questions

- Who is remembered by society after death?
 - “Postmortem collective memory”
- Are there prototypical patterns of postmortem collective memory?
- Are certain kinds of people remembered in certain ways?
- Are dead people remembered differently in news vs. social media?



Philip Seymour Hoffman † 2014



Amy Winehouse † 2011

5



6

Why should we care about postmortem collective memory?

Fact: Humans care a lot about being remembered after death



Ara Pacis, Rome
[\[Wikipedia\]](#)



Damnatio memoriae
[\[Wikipedia\]](#)

7

An ADA approach

Let's use lots of data and count stuff!

- **Detect names** of dead people in big corpus of news and social media
- Build time series of name **counts**
- Analyze the **shape of time series**
- **Correlate** shapes with biographic info about dead people from knowledge base



Stuffed Count von Count counting stuff

8

Part 1: Getting the data

The raw data: Spinn3r

- "Spinn3r provides APIs for **social media**, **weblogs**, **news**, video, and live web content to our customers in any language and in large volumes." (Source: spinn3r.com)
- Firehose stored to disk
- Several billions of documents
- 40 terabytes

Postmortem memory of public figures in news and social media

Robert West*, Jure Leskovec*, and Christopher Potts*
*School of Computer and Communication Sciences, Ecole Polytechnique Fédérale de Lausanne, 1015 Lausanne, Switzerland; *Department of Computer Science, Stanford University, Stanford, CA 94305; and *Department of Linguistics, Stanford University, Stanford, CA 94305
Edited by Henry L. Roediger III, Washington University in St. Louis, MO, and approved June 24, 2021 (received for review April 1, 2021)

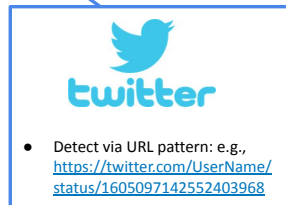
ada #12 Scaling to massive data

9

10

Detecting news and social media in Spinn3r

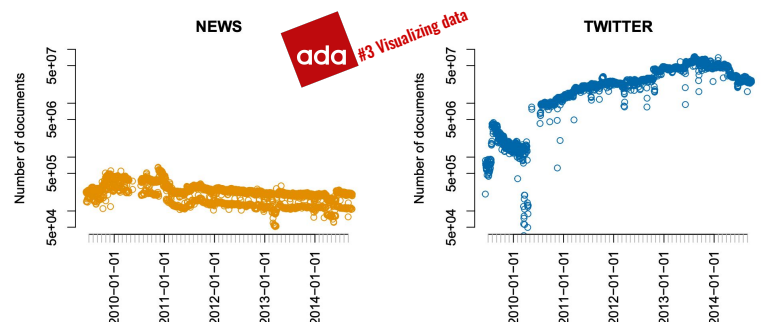
- Found a [list](#) of all 151K online news articles about Osama bin Laden's killing (2 May 2011) indexed by Google News
- Assume that every relevant news outlet had reported on bin Laden's death, and that Google News is reasonably complete
- News defined as documents from the 6,608 Web domains appearing in the "bin Laden list"



ada #2 Handling data

11

Data volume: Number of docs per day



Detecting mentions of people names

- Easy: unambiguous names
 - e.g., "Amy Winehouse"
- Hard: ambiguous names
 - e.g., "Michael Jackson"

- Wikipedia-based solution:
- Given: a name X (e.g., "Michael Jackson")
 - Consider all N links in English Wikipedia with X as anchor text (cf. examples on the right)
 - Build distribution over the N link targets
 - If there is a target T to which $\geq 90\%$ of all links point: consider X "sufficiently unambiguous" and assume that all mentions of X in Spinn3r refer to T
 - Else: consider X "too ambiguous" and ignore it in the analysis

Michael Jackson (disambiguation)

Michael Jackson (1958–2009) was an American singer, songwriter and dancer known as the "King of Pop".

Michael Jackson, Mike Jackson, or Mick Jackson may also refer to:

- People
 - Michael Jackson (disambiguation) (1958–2009), American radio talk show host, A&R, and A&R, Los Angeles
 - Michael Jackson (producer) (1962–2017), film director, producer, actor and writer
 - Michael Jackson (actor) (born 1972), British actor, known for The Underground
 - Mike Jackson (producer) (born 1962), British film and TV director, known for The Bridge
 - Michael J. Jackson (born 1948), English actor from Liverpool, best known for his role in The Godfather
 - Michael Jackson (actor) (born 1972), Canadian actor
 - Mike Jackson (film producer) (born 1972), American film producer and talent manager
 - Michael S. Jackson (born 1947), American playwright, composer, and lyricist
 - Mike Jackson (musician) (1988–2018), American jazz pianist and composer
 - Mike Jackson (Australian politician) (born 1946), Australian rock instrumentalist, singer and children's entertainer
 - Michael Jackson (singer) (born 1958), English singer-songwriter
 - Michael Jackson (politician) (born 1964), Australian politician, born Michael Gregory Jackson
 - Mike and Michael Jackson, Australian multi-instrumental duo
 - Michael Jackson (English organist) (born 1964), British organist with the Henry Wood Hall Organ Festival
 - Dr. Mike Jackson, birth name Michael Winehouse Jackson (born 1970), American rapper
 - Michael Lee Jackson, guitarist
 - Mike Jackson, bassist with British band Love Affair (1980s)
- Military and militia
 - Michael Jackson (British soldier) (1734–1825), soldier from Massachusetts, recruited at Bunker Hill
 - Mike Jackson (British Army officer) (born 1944), former head of the British

Recruiting the army of the dead:

	All	Included
Age		
N/A	10%	6%
1st quartile	68	64
Mean	76	74
Median	80	77
3rd quartile	88	87
Gender		
N/A	27%	7%
Female	16%	17%
Male	84%	83%
Manner of death		
N/A	76%	69%
Natural	85%	86%
Unnatural	15%	14%
Language		
N/A	45%	27%
Anglophone	60%	82%
Non-anglophone	40%	18%
Notability type		
N/A	1%	0%
Arts	40%	60%
Sports	14%	14%
Leadership	11%	14%
Known for death	26%	16%
General fame	7%	4%
Academia/engineering	2%	2%
Count	3340	2362

- Freebase, a once-popular knowledge graph
 - Now defunct, but still [available](#) 
- Contains information about >3M people
 - e.g., Philip Seymour Hoffman = /m/02qgqt
 - More info about some, less about others
- Start from 33K people with death date 2009–2014
 - Further filtering → 2,362 people
- Extract **biographic information** from Freebase

ada #2 Handling data #13 Handling network data

Tools used

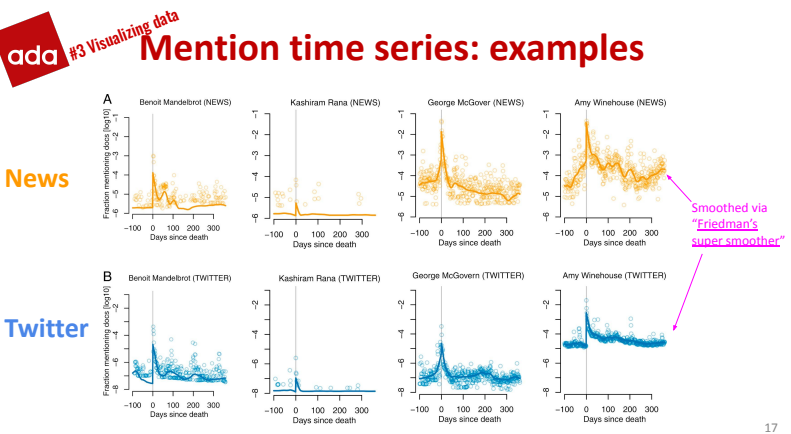
- Counting names in Spinn3r dump: Hadoop (Java)
- Extracting info from Freebase: Python, Perl
- Once data was small enough: R ([script](#), [repo](#))
 - Statistical analyses
 - Plotting
 - Read data as CSV once (minutes), then serialized to binary format and deserialized in later runs (seconds)

Our protagonist: Mention time series

- t : relative time t , counting days since death
 - $t = 0$: day of death
- $S_i(t)$: fraction of documents in which person i was mentioned, out of all documents published on day t
 - For mention time series, consider logarithms:
 $\log_{10} S_i(t)$

ada #1–13 ADA loves logs!

Mention time series: examples

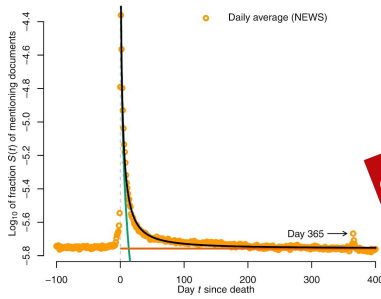


Part 2:

The shape of postmortem memory

Average mention time series

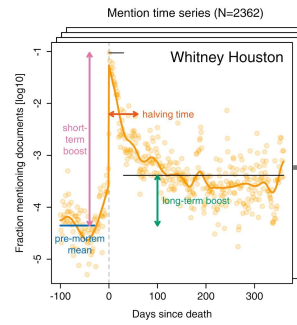
News



$$\left(\prod_{i=1}^n a_i\right)^{\frac{1}{n}} = \sqrt[n]{a_1 a_2 \dots a_n} = \exp\left(\frac{1}{n} \sum_{i=1}^n \ln a_i\right)$$

19

Curve characteristics



Pre-mortem mean: arithmetic mean of days 360 through 30 before death

Short-term boost: maximum of days 0 through 29 after death, minus the premortem mean

Long-term boost: arithmetic mean of days 30 through 360 after death, minus the premortem mean

Halving time: number of days required to accumulate half of the total area between the postmortem curve (including the day of death) and the minimum postmortem value

Median over people: 1.98
95% CI [1.90, 2.03]

Median over people: 0.00055
95% CI [-0.00091, 0.0017]

ada #4 Describing data

20

Commercial break

WIKIPEDIA The Free Encyclopedia

Adenosine deaminase

From Wikipedia, the free encyclopedia

Adenosine deaminase (also known as **adenosine aminohydrolase**, or **ADA**) is an enzyme (EC 3.5.4.4^(?)) involved in **purine metabolism**. It is needed for the breakdown of **adenosine** from food and for the turnover of **nucleic acids** in tissues.

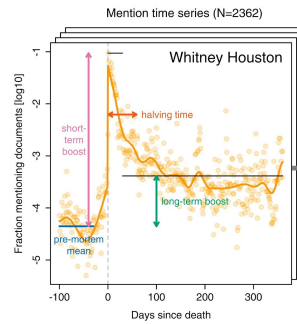
Its primary function in humans is the development and maintenance of the immune system.^[6] However, the full physiological role of ADA is not yet completely understood.^[6]

Structure [[edit](#)]

ADA exists in both small form (as a monomer) and large form (as a dimer-complex).^[6] In the monomer form, the enzyme is a polypeptide chain,^[7] folded into eight strands of parallel α/β barrels, which surround a central deep pocket that is the active site.^[6] In addition to the eight central β-barrels and eight peripheral α-helices, ADA also contains five additional helices: residues 19-76 fold into three helices, located between β1 and α1 folds; and two antiparallel

21

Are there prototypical curve shapes?



- Each curve one data point
- Represented via its 4 curve characteristics
 - ≈ manual dimensionality reduction
 - Values standardized via z-scores
- Cluster via *k*-means

Q: How to find the best number *k* of clusters?

ada #8 Applied ML
#9 Unsupervised learning

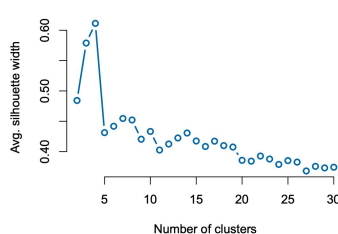
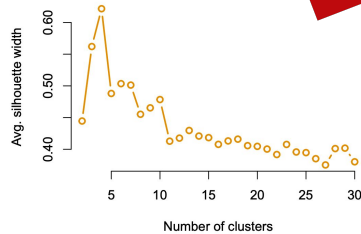
22

Average silhouette width!

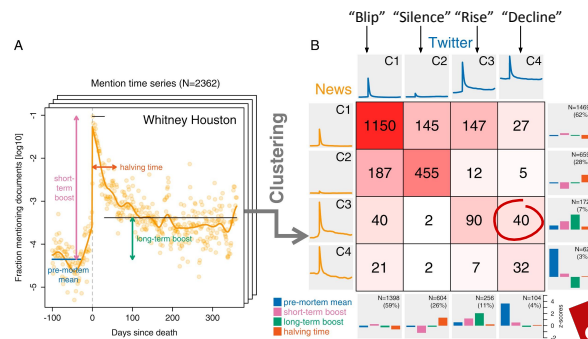
News

ada #9 Unsupervised learning

Twitter



Cluster analysis



ada #4 Describing data

24

A first stab

Part 3:

Biographic correlates of postmortem memory

- Measure correlation coefficient between outcome (e.g., short-term memory boost) and each biographic properties (e.g., gender)
- Higher correlation $\Rightarrow$ higher outcome for the respective group of people (e.g., women)



Age	N/A
1st quartile	
Mean	
Median	
3rd quartile	
Gender	N/A
Female	
Male	
Manner of death	N/A
Natural	
Unnatural	
Language	N/A
Anglophone	
Non-anglophone	
Notability type	N/A
Arts	
Sports	
Leadership	
Known for death	
General fame	
Academia/engineering	
Count	

25

26

Problem: Biographic properties are correlated

E.g., leaders (politicians, CEOs, etc.) are

- more likely to have died old,
 - more likely to have died of a natural death,
 - more likely to be men,
- compared to artists



Regression analysis allows us to compare averages across subgroups of the data while accounting for correlations among averaged values!

ada #5 Regression analysis

Linear regression

Avg. outcome for "baseline persona": male anglophone artist of median premortem popularity who died a natural death at age 70–79

$$y_i = \beta_0 + \beta_1 \text{premortem_mention_freq}_i + \beta_2 \text{age_at_death}_i + \beta_3 \text{manner_of_death}_i + \beta_4 \text{notability_type}_i + \beta_5 \text{language}_i + \beta_6 \text{gender}_i$$

Outcome for person i :

- short-term boost or
- long-term boost

Rank-transformed, then linearly scaled/shifted to $[-0.5, 0.5]$; i.e., median has value 0

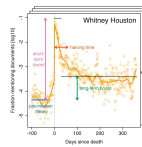
8 discrete levels (dummy-coded): 20–29, 30–39, ..., 70–79, 80–89, 90–99

2 levels: natural, unnatural

6 levels: arts, sports, leadership, known for death, general fame, academia/engineering

3 levels: anglophone, non-anglophone, unknown

2 levels: male, female



27

28

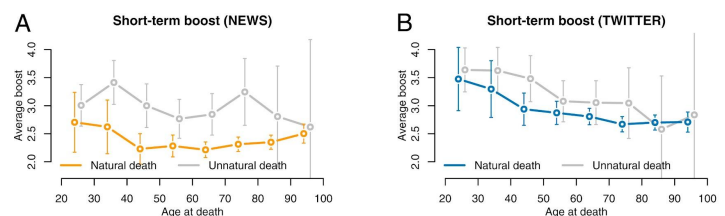
Linear regression results

	Short-term boost (news)	Short-term boost (Twitter)	Long-term boost (news)	Long-term boost (Twitter)
(Intercept)	2.322 (0.063)***	2.670 (0.067)***	0.088 (0.014)***	0.095 (0.015)***
Premortem mean (relative rank)	0.804 (0.093)***	0.948 (0.100)***	0.031 (0.020)	0.086 (0.022)***
Manner of death: unnatural	0.618 (0.095)***	0.282 (0.100)**	0.097 (0.021)***	0.106 (0.022)***
Language: non-anglophone	-0.316 (0.074)***	-0.116 (0.078)	-0.061 (0.016)***	-0.037 (0.017)*
Language: unknown	-0.446 (0.086)***	-0.325 (0.091)***	-0.079 (0.019)***	-0.081 (0.020)***
Gender: female	0.083 (0.072)	-0.034 (0.076)	0.034 (0.016)*	0.006 (0.017)
Notability type: academia/engineering	0.181 (0.197)	0.340 (0.208)	-0.032 (0.043)	0.023 (0.046)
Notability type: general fame	0.070 (0.124)	0.132 (0.131)	-0.010 (0.027)	-0.008 (0.029)
Notability type: known for death	-0.107 (0.099)	-0.088 (0.106)	-0.021 (0.022)	0.008 (0.023)
Notability type: leadership	0.152 (0.083)	0.113 (0.087)	-0.058 (0.018)**	-0.040 (0.019)*
Notability type: sports	0.049 (0.083)	0.072 (0.088)	-0.034 (0.018)	-0.034 (0.020)
Age: 20–29	0.162 (0.170)	0.718 (0.180)***	0.060 (0.037)	0.192 (0.040)***
Age: 30–39	0.400 (0.167)*	0.649 (0.177)***	0.028 (0.037)	0.118 (0.039)**
Age: 40–49	-0.046 (0.126)	0.351 (0.133)**	-0.001 (0.028)	0.100 (0.030)***
Age: 50–59	-0.075 (0.099)	0.181 (0.104)	-0.058 (0.022)**	-0.024 (0.023)
Age: 60–69	-0.109 (0.082)	0.130 (0.086)	-0.050 (0.018)**	-0.025 (0.019)
Age: 80–89	0.022 (0.078)	0.021 (0.082)	-0.018 (0.017)	-0.013 (0.018)
Age: 90–99	0.174 (0.098)	0.034 (0.103)	-0.011 (0.021)	-0.024 (0.023)
R ²	0.213	0.192	0.123	0.178
Adj. R ²	0.197	0.176	0.106	0.161
No. obs.	870	870	870	870
RMSE	0.772	0.815	0.169	0.181

S.E.s of coefficients are in parentheses. ***P < 0.001, **P < 0.01, and *P < 0.05.

29

Age at death vs. postmortem memory



News plays two simultaneous roles (more so than Twitter):

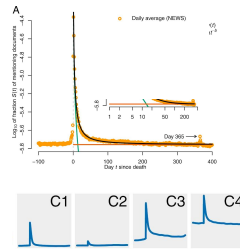
- Catering to public curiosity stirred by a young or unnatural death
- But also when old person or accomplished leader dies

30

Part 4: Discussion

Summary: The shape of postmortem memory

- Sharp **pulse** of media attention with death:
 - Median: +9,400% in news, +28,000% on Twitter
- Then sharp **drop** (around 1 month long) toward premortem level
- **Cluster analysis** revealed a set of four prototypical memory patterns: “Blip”, “silence”, “rise”, “decline”
- Same patterns in news and Twitter; **same person** tends to fall into the **same cluster** across the two media



31

32

Summary: Biographic correlates

- Notability types: all regression coefficients for long-term boosts negative
⇒ All types have lower average long-term boost than default type (artists)
⇒ **Artists more present in collective memory**
- **Low R^2** (10–20%): human lives/legacies rich, hard to model
 - But all **model fits highly significant** (F -statistic, p -value)
 - **Effects** not only significant, but also **large**: e.g., short-term boost (on linear scale) for unnatural vs. natural death: 4x in news, 2x on Twitter
- **Largest boost:**
 - **Premortem popular anglophones who died a young, unnatural death**
 - Long-term boosts **largest for artists, smallest for leaders**

33

